

# Chapter 5

## Discrete Random Variables

In the previous chapters, we learned how to describe our data using summary statistics. Now, we take the next leap: to build models that explain the underlying processes generating this data.

Think about it – when we take measurements, they rarely come out exactly the same every time. There's often variability we can't fully control or even understand.

To account for this inherent unpredictability, we'll start treating our measurements as **random variables (RVs)**. These are variables whose values have a component of randomness, a kind of “statistical noise.”

- We often use what we call random variables to describe the important aspects of the outcomes of experiments.
- In this chapter we introduce two important types of random variables — **discrete** random variables and **continuous** random variables
- and learn how to find probabilities concerning discrete random variables.
- As one application, we will begin to see how to use probabilities concerning discrete random variables to make statistical inferences about populations.

- 5.1 Two Types of Random Variables**
- 5.2 Discrete Probability Distributions**
- 5.3 The Binomial Distribution**

## Section 5.1

# Two Types of Random Variables

- A **random variable** is a variable that assumes numerical values that are determined by the outcome of an experiment, where one and only one numerical value is assigned to each experimental outcome.

(the value of the variable is determined by the experiment. One experiment for one value)

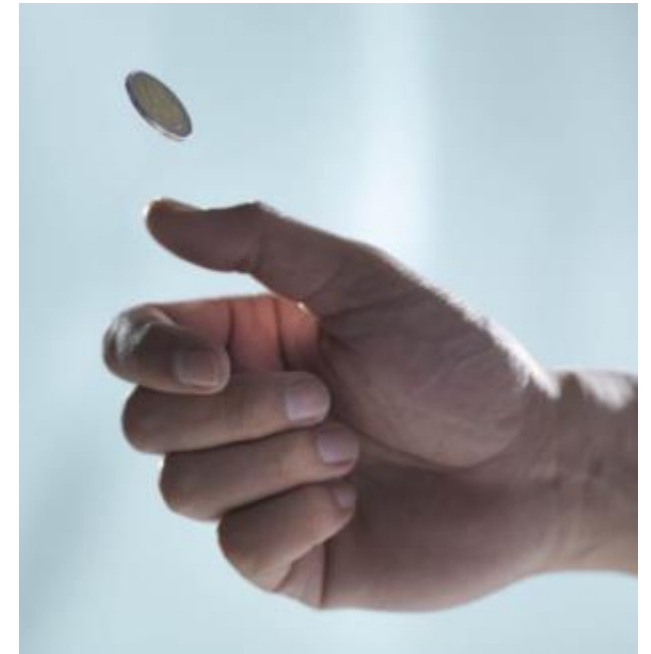
- Before an experiment is carried out, its outcome is **uncertain**. It follows that, because a random variable assigns a number to each experimental outcome, a random variable can be thought of as representing ***an uncertain numerical outcome***.

(before the experiment, we do not know the value of the variable).

- Loosely speaking, a **random variable** is a (numerical) variable whose **outcome is uncertain**. Typically, random variables arise by assigning a value to the outcome of some “experiment”.
- **Definition:** More formally, the random variable  $X$  can be defined as a function which assigns a number to each outcome in the **sample space**.

## Example

- Consider a random experiment in which a coin is tossed three times. Let  $X$  be the number of heads. Let  $H$  represent the outcome of a head and  $T$  the outcome of a tail.
- The possible outcomes for such an experiment:  
**TTT, TTH, THT, THH, HTT, HTH, HHT, HHH**
- Thus the possible values of  $X$  (number of heads) are 0, 1, 2, 3.
- From the definition of a random variable,  $X$  as defined in this experiment, is a **random variable**.



# Two Types of Random Variables

- **Discrete random variable:** Possible values can be counted or listed
  - ▣ For example, the number of TV sets sold at the store in one day. Here  $x$  could be 0, 1, 2, 3, 4 and so forth.
- **Continuous random variable:** May assume any numerical value in one or more intervals
  - For example, the waiting time for a credit card authorization, the interest rate charged on a business loan, the speed of a vehicle.

## Example: Two Types of Random Variables

| Question                    | Random Variable $x$                                                                                          | Type       |
|-----------------------------|--------------------------------------------------------------------------------------------------------------|------------|
| Family size                 | $x$ = Number of people in family reported on tax return                                                      | Discrete   |
| Distance from home to store | $x$ = Distance in miles from home to a store                                                                 | Continuous |
| Own dog or cat              | $x$ = 1 if own no pet;<br>= 2 if own dog(s) only;<br>= 3 if own cat(s) only;<br>= 4 if own dog(s) and cat(s) | Discrete   |

## **Section 5.2**

# **Discrete Probability Distributions**

The **probability distribution** of a discrete random variable is a table, graph, or formula that gives the probability associated with each possible value that the variable can assume.

**Notation:** Denote the values of the random variable by  $x$  and the value's associated probability by  $p(x)$ .

## Properties

1. For any value  $x$  of the random variable,  $p(x) \geq 0$
2. The probabilities of all the events in the sample space must sum to 1, that is,

$$\sum_{\text{all } x} p(x) = 1$$

## Example 5.2: Number of Radios (Sold at Sound City in a Week)

- Let  $x$  be the random variable of the number of radios sold per week
  - $x$  has values  $x = 0, 1, 2, 3, 4, 5$
- Given sales history over past 100 weeks
  - Let  $f$  be the number of weeks (of the past 100) during which  $x$  number of radios were sold
  - Records tell us  $f(x)$ 
    - $f(0) = 3$       No radios have been sold in 3 of the weeks
    - $f(1) = 20$       One radio has been sold in 20 of the weeks
    - $f(2) = 50$       Two radios have been sold in 50 of the weeks
    - $f(3) = 20$       Three radios have been sold in 20 of the weeks
    - $f(4) = 5$       Four radios have been sold in 4 of the weeks
    - $f(5) = 2$       Five radios have been sold in 2 of the weeks

## Checking the conditions

- $f(0) = 3$     there are 3 weeks selling 0 radio every week
- $f(1) = 20$     there are 20 weeks selling 1 radio every week
- $f(2) = 50$     there are 50 weeks selling 2 radios every week
- $f(3) = 20$     there are 20 weeks selling 3 radios every week
- $f(4) = 5$     there are 5 weeks selling 4 radios every week
- $f(5) = 2$     there are 2 weeks selling 5 radios every week

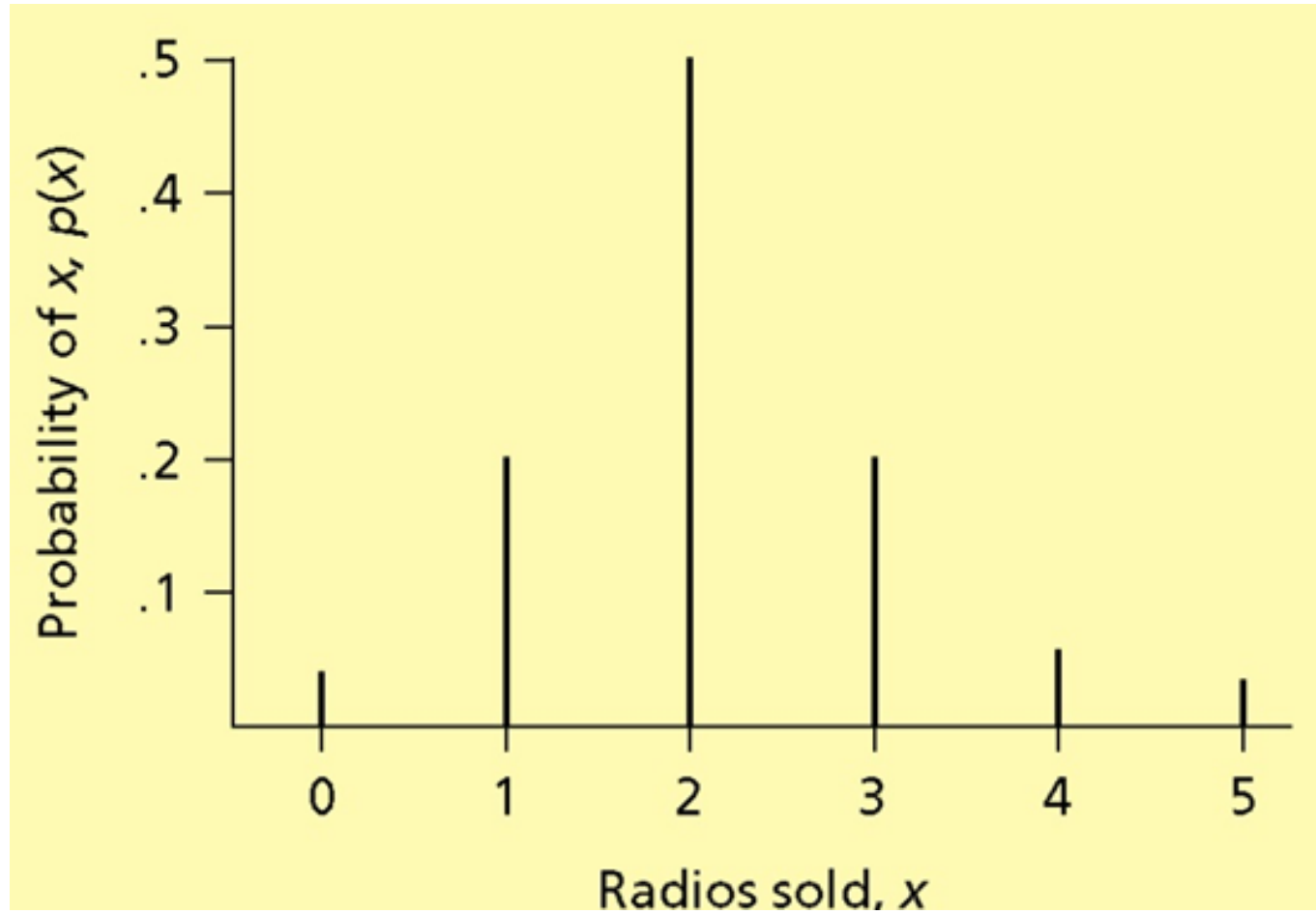
# Frequency distribution of sales history over past 100 weeks

**TABLE 5.1**

| Radios, $x$ | Frequency              | Relative Frequency | Probability, $p(x)$       |
|-------------|------------------------|--------------------|---------------------------|
| 0           | $f(0) = 3$             | $3/100 = 0.03$     | $p(0) = 0.03$             |
| 1           | $f(1) = 20$            | $20/100 = 0.20$    | $p(1) = 0.20$             |
| 2           | $f(2) = 50$            | 0.50               | $p(2) = 0.50$             |
| 3           | $f(3) = 20$            | 0.20               | $p(3) = 0.20$             |
| 4           | $f(4) = 5$             | 0.05               | $p(4) = 0.05$             |
| 5           | $f(5) = \underline{2}$ | <u>0.02</u>        | $p(5) = \underline{0.02}$ |
|             | 100                    | 1.00               | 1.00                      |

- Interpret the relative frequencies as probabilities
  - So for any value  $x$ ,  $f(x)/n = p(x)$
  - Assuming that sales remain stable over time

# Probability distribution



- What is the chance that two radios will be sold in a week?

- $P(x = 2) = 0.50$

- What is the chance that fewer than two radios will be sold in a week?

- $P(x < 2) = p(x = 0 \text{ or } x = 1)$   
 $= p(x = 0) + p(x = 1)$   
 $= 0.03 + 0.20 = 0.23$

Using the addition rule for the mutually exclusive values of the random variable.

- What is the chance that three or more radios will be sold in a week?

- $P(x \geq 3) = p(x = 3, 4, \text{ or } 5)$   
 $= p(x = 3) + p(x = 4) + p(x = 5)$   
 $= 0.20 + 0.05 + 0.02 = 0.27$

## The Mean, or Expected Value, of a Discrete Random Variable

In above example, total number of radios sold in the past 100 weeks can be calculated as below table

| Frequency   | Number of radios sold |
|-------------|-----------------------|
| $f(0) = 3$  | $0 \times 3 = 0$      |
| $f(1) = 20$ | $1 \times 20 = 20$    |
| $f(2) = 50$ | $2 \times 50 = 100$   |
| $f(3) = 20$ | $3 \times 20 = 60$    |
| $f(4) = 5$  | $4 \times 5 = 20$     |
| $f(5) = 2$  | $5 \times 2 = 10$     |
| 100         | 210                   |

By average (the mean), the number of radios sold in one week is

$$\mu_x = \frac{210}{100} = \frac{\sum_{all\ x} f(x)x}{n}$$

or

$$\mu_x = \sum_{all\ x} \frac{f(x)}{n} x = \sum_{all\ x} p(x) x$$

## The Mean, or Expected Value, of a Discrete Random Variable

- Suppose that the experiment described by a random variable  $x$  is repeated an indefinitely large number of times.
- If the values of the random variable  $x$  observed on the repetitions are recorded, we would obtain the population of all possible observed values of the random variable  $x$ .
- This population has a **mean**, which we denote as  $\mu_x$  and which we sometimes call the **expected value** of  $x$ . In order to calculate  $\mu_x$ , we multiply each value of  $x$  by its probability  $p(x)$  and then sum the resulting products over all possible values of  $x$ .

The mean, or expected value, of a discrete random variable  $x$  is

$$\mu_x = \sum_{\text{all } x} xp(x)$$

$\mu_x$  is the value expected to occur in the long run and on average

$p(x)$ : the weight. If the chance is bigger, the weight is higher.

### Example 5.3: Number of Radios Sold at Sound City in a Week

- How many radios should be expected to be sold in a week?
- Calculate the **expected value** of the number of radios sold,  $m_X$

| Radios, $x$ | Probability, $p(x)$             | $xp(x)$                                  |
|-------------|---------------------------------|------------------------------------------|
| 0           | $p(0) = 0.03$                   | $0 \times 0.03 = 0.00$                   |
| 1           | $p(1) = 0.20$                   | $1 \times 0.20 = 0.20$                   |
| 2           | $p(2) = 0.50$                   | $2 \times 0.50 = 1.00$                   |
| 3           | $p(3) = 0.20$                   | $3 \times 0.20 = 0.60$                   |
| 4           | $p(4) = 0.05$                   | $4 \times 0.05 = 0.20$                   |
| 5           | <u><math>p(5) = 0.02</math></u> | <u><math>5 \times 0.02 = 0.10</math></u> |
|             | 1.00                            | <b>2.10</b>                              |

- On **average**, expect to sell 2.1 radios per week

## Example 5.4: The Life Insurance Case: Setting a Policy Premium

An insurance company sells a \$20,000 whole life insurance policy for an annual premium of \$300. Actuarial tables show that a person who would be sold such a policy with this premium has a 0.001 probability of death during a year. Let  $x$  be a **random variable** representing the insurance company's **profit** made on one of these policies during a year. The **probability distribution of  $x$**  is

**$x$ , Profit**

\$300 (if the policyholder lives)

$\$300 - \$20,000 = -\$19,700$

(a \$19,700 loss if the policyholder dies)

**$p(x)$ , Probability of  $x$**

.999

.001

## Checking the conditions

- The insurance company sells insurance to a person at \$300 per year.
- If the person does not die, the insurance company will get \$300 for that year. So the profit of the company is \$300.
- If the person dies in the year, the company will give the person \$20,000. So the profit of the insurance company is negative \$17,000.
- The possibility (chance) the person dies is 0.001.
- The money the insurance made is a random variable and is denoted as  $x$
- The **probability distribution of  $x$**  is

**$x$ , Profit**

\$300 (if the policyholder lives)

$\$300 - \$20,000 = -\$19,700$

(a \$19,700 loss if the policyholder dies)

**$p(x)$ , Probability of  $x$**

.999

.001

The expected value of  $x$  (expected profit per year) is

$$\mu_x = \$300 \times 0.999 + (-\$19,700) \times 0.001 = \$280.0$$

This says that if the insurance company sells a very large number of these policies, it will average a profit of \$280 per policy per year.

Because insurance companies actually do sell large numbers of policies, it is reasonable for these companies to make profitability decisions based on expected values.

In general, it is reasonable to base decisions on an expected value if we perform the experiment related to the decision (for example, if we sell the life insurance policy) **many times**. If we do not (for instance, if we perform the experiment only once), then it may not be a good idea to base decisions on the expected value.

For example, if you are selling only one policy, you will not receive the \$280. You will either gain \$300 (with probability 0.999) or you will lose \$19,700 (with probability 0.001). Although the chance of losing \$19,700 is very small, no company would risk such a loss when the potential gain is only \$300.

# Variance and Standard Deviation of a Discrete Random Variable

- The **variance** of a discrete random variable is:

$$\sigma_x^2 = \sum_{\text{All } x} (x - \mu_x)^2 p(x)$$

The **variance** is the average of the squared deviations of the different values of the random variable from the expected value.

- The **standard deviation** is the square root of the variance

$$\sigma_x = \sqrt{\sigma_x^2}$$

The variance and standard deviation measure the spread of the values of the random variable from their expected value (the mean).

### Example 5.5: Number of Radios (Sold at Sound City in a Week)

Recall **example 5.2**. Calculate the **variance and standard deviation** of the number of radios sold at Sound City in a week

| Radios, $x$ | Frequency              | Probability, $p(x)$       | $(x - \mu_x)^2 p(x)$                      |
|-------------|------------------------|---------------------------|-------------------------------------------|
| 0           | $f(0) = 3$             | $p(0) = 0.03$             | $(0 - 2.1)^2 (0.03) = 0.1323$             |
| 1           | $f(1) = 20$            | $p(1) = 0.20$             | $(1 - 2.1)^2 (0.20) = 0.2420$             |
| 2           | $f(2) = 50$            | $p(2) = 0.50$             | $(2 - 2.1)^2 (0.50) = 0.0050$             |
| 3           | $f(3) = 20$            | $p(3) = 0.20$             | $(3 - 2.1)^2 (0.20) = 0.1620$             |
| 4           | $f(4) = 5$             | $p(4) = 0.05$             | $(4 - 2.1)^2 (0.05) = 0.1805$             |
| 5           | $f(5) = \underline{2}$ | $p(5) = \underline{0.02}$ | $(5 - 2.1)^2 (0.02) = \underline{0.1682}$ |
|             | 100                    | 1.00                      | 0.8900                                    |

Variance

$$\sigma_x^2 = 0.8900$$

Standard deviation

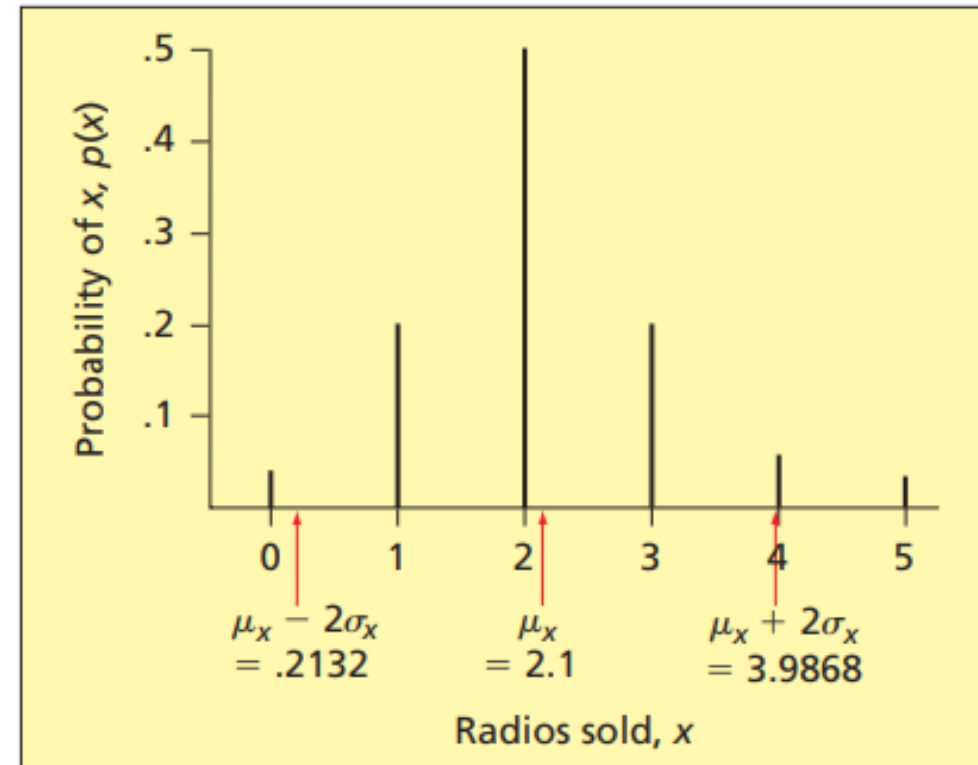
$$\sigma_x = \sqrt{0.8900} = 0.9434$$

We can consider the probability that a random variable  $x$  will be in a region. For example, consider the probability distribution in Table 5.1 of  $x$ , the number of radios sold in a week at Sound City. If (for instance) we wish to find the probability that  $x$  will be within (plus or minus) 2 standard deviations of  $\mu_x$ , then we need to find the probability that  $x$  will lie in the interval

$$[\mu_x \pm 2\sigma_x] = [0.2132, 3.9868]$$

As illustrated in Figure 5.2, there are three values of that lie in the interval  $[0.2132, 3.9868]$ . Therefore, the probability that  $x$  will lie in the interval  $[0.2132, 3.9868]$  is the probability that  $x$  will equal 1 or 2 or 3, which is  $p(1) + p(2) + p(3) = 0.20 + 0.50 + 0.20 = 0.90$ . This says that in 90 percent of all weeks, the number of radios sold at Sound City will be within (plus or minus) two standard deviations of the mean weekly sales of the radio at Sound City.

Figure 5.2 The Interval  $[\mu_x \pm 2\sigma_x]$  for the Probability Distribution Describing Radio Sales (see Table 5.1)

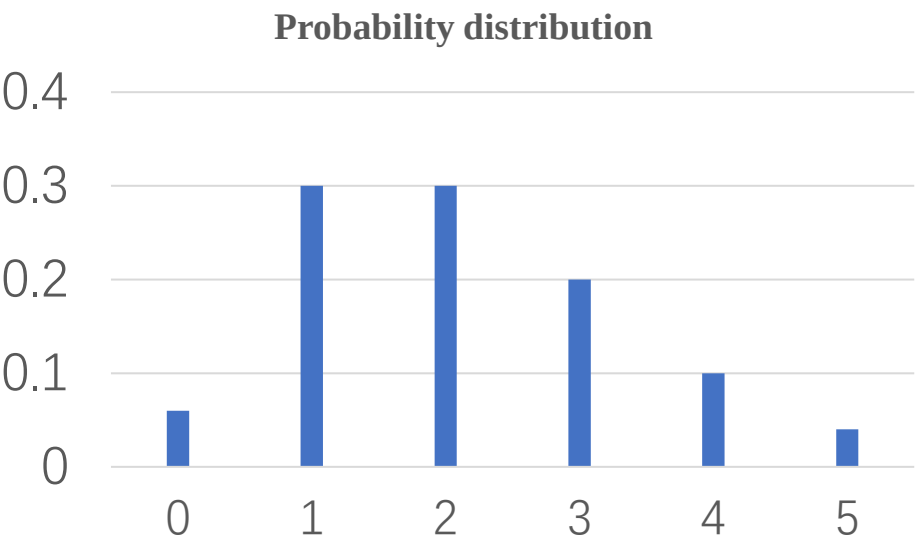


## Exercises (do this by yourself)

**5.8** Recall from Example 5.5 that Sound City also sells the ClearTone-400 satellite car radio. For this radio, historical sales records over the last 100 weeks show 6 weeks with no radios sold, 30 weeks with one radio sold, 30 weeks with two radios sold, 20 weeks with three radios sold, 10 weeks with four radios sold, and 4 weeks with five radios sold. Estimate and write out the probability distribution of  $x$ , the number of ClearTone-400 radios sold at Sound City during a week.

# Solution

| Radios, $x$ | Frequency              | Probability, $p(x)$ |
|-------------|------------------------|---------------------|
| 0           | $f(0) = 6$             | $6/100 = 0.06$      |
| 1           | $f(1) = 30$            | $30/100 = 0.30$     |
| 2           | $f(2) = 30$            | 0.30                |
| 3           | $f(3) = 20$            | 0.20                |
| 4           | $f(4) = 10$            | 0.10                |
| 5           | $f(5) = \underline{4}$ | $\underline{0.04}$  |
|             | 100                    | 1.00                |



You can use either the table (the left) or the graph (right) to describe the probability distribution

## Exercises (do this by yourself)

**5.9** Use the estimated probability distribution in Exercise 5.8 to calculate  $\mu_x$ ,  $\sigma_x^2$ , and  $\sigma_x$ .

### Solution

$$\mu_x = 0 \times 0.06 + 1 \times 0.3 + 2 \times 0.3 + 3 \times 0.2 + 4 \times 0.1 + 5 \times 0.04 = 2.1$$

$$\sigma_x^2 = (0 - 2.1)^2(0.06) + (1 - 2.1)^2(0.3) + (2 - 2.1)^2(0.3) + (3 - 2.1)^2(0.2) + (4 - 2.1)^2(0.1) + (5 - 2.1)^2(0.04) = 1.49$$

$$\sigma_x = \sqrt{1.49} = 1.22$$

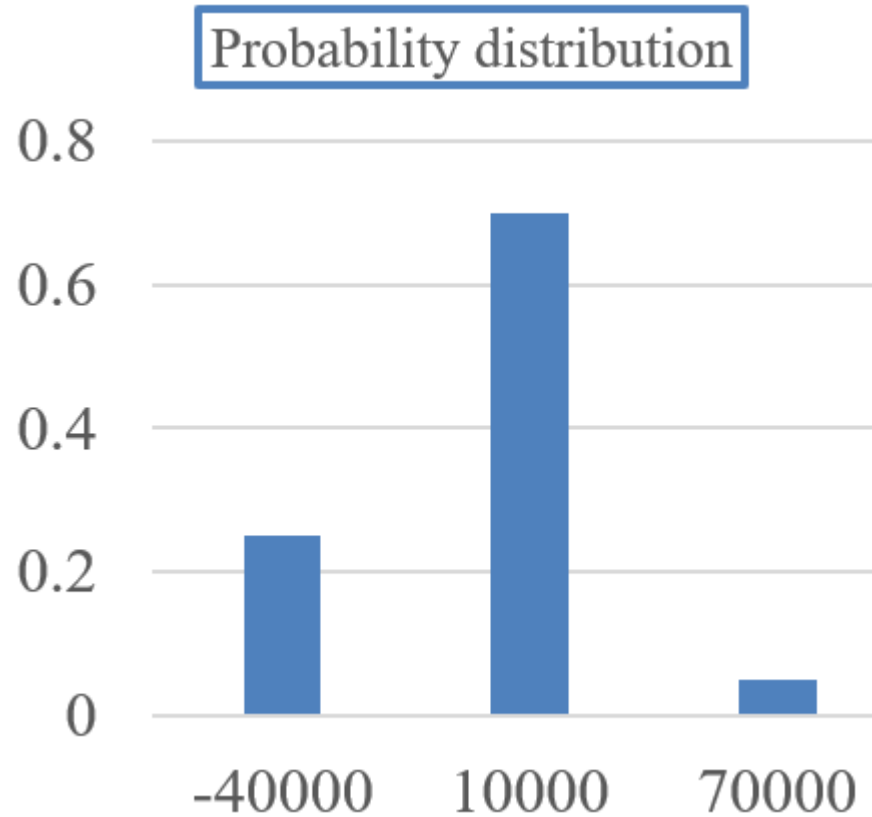
## Exercises (do this by yourself, will be explained in the class only if time permits)

**5.11** The following table summarizes investment outcomes and corresponding probabilities for a particular oil well.

- (a) Graph  $p(x)$ ; that is, graph the probability distribution of  $x$ .
- (b) Find the expected monetary outcome. Mark this value on your graph of part (a) Then interpret this value.
- (c) Calculate the standard deviation of  $x$ .

| $x = \text{the outcome in \$}$ | $p(x)$ |
|--------------------------------|--------|
| –\$40,000 (no oil)             | .25    |
| 10,000 (some oil)              | .7     |
| 70,000 (much oil)              | .05    |

## Solution



(b) Expected monetary outcome

$$\mu_x = (-40,000) \times 0.25 + (10,000) \times 0.7 + (70,000) \times 0.05 = 500$$

## Exercises (will be explained in the class)

- 5.15** Five thousand raffle tickets are to be sold at \$10 each to benefit a local community group. The prizes, the number of each prize to be given away, and the dollar value of winnings for each prize are as follows:  Raffle

| Prize                | Number to Be Given Away | Dollar Value |
|----------------------|-------------------------|--------------|
| Automobile           | 1                       | \$20,000     |
| Entertainment center | 2                       | 3,000 each   |
| DVD recorder         | 5                       | 400 each     |
| Gift certificate     | 50                      | 20 each      |

If you buy one ticket, calculate your expected winnings. (Form the probability distribution of  $x$  = your dollar winnings, and remember to subtract the cost of your ticket.)

# Solution

| Prize, $x$           | Frequency                  | Probability, $p(x)$ |
|----------------------|----------------------------|---------------------|
| None                 | $f(-10) = 4942$            | $10/5,000 = 0.9884$ |
| Automobile           | $f(19,990) = 1$            | $1/5,000 = 0.0002$  |
| Entertainment center | $f(2,990) = 2$             | $0.0004$            |
| DVD recorder         | $f(390) = 5$               | $0.001$             |
| Gift certificate     | $f(10) = \frac{50}{5,000}$ | $\frac{0.01}{1.00}$ |

The average winning is  $\mu_x = (-10) \times 0.9884 + (19,990) \times 0.0002 + (2,990) \times 0.0004 + (390) \times 0.001 + (10) \times 0.01 = -4.2$

## Exercises (do this by yourself)

**5.16** A survey conducted by a song rating service finds that the percentages of listeners familiar with *Poker Face* by Lady Gaga who would give the song ratings of 5, 4, 3, 2, and 1 are, respectively, 43 percent, 21 percent, 22 percent, 7 percent, and 7 percent. Assign the numerical values 1, 2, 3, 4, and 5 to the (qualitative) ratings 1, 2, 3, 4, and 5 and find an estimate of the probability distribution of  $x$  this song's rating by a randomly selected listener who is familiar with the song.

## Solution

| Rate, $x$ | Probability, $p(x)$ |
|-----------|---------------------|
|-----------|---------------------|

|   |               |
|---|---------------|
| 1 | $f(5) = 0.43$ |
|---|---------------|

|   |               |
|---|---------------|
| 2 | $f(4) = 0.21$ |
|---|---------------|

|   |               |
|---|---------------|
| 3 | $f(3) = 0.22$ |
|---|---------------|

|   |               |
|---|---------------|
| 4 | $f(2) = 0.07$ |
|---|---------------|

|   |               |
|---|---------------|
| 5 | $f(1) = 0.07$ |
|---|---------------|

**5.17** In Exercise 5.16,

(a) Find the **expected value** of the estimated probability distribution.

(b) Explain the meaning of this expected value in terms of all possible *Poker Face* listeners.

### Solution

$$\begin{aligned}\mu_x &= 5 \times 0.43 + 4 \times 0.21 + 3 \times 0.22 + 2 \\ &\times 0.07 + 1 \times 0.07 = 3.86\end{aligned}$$

By average, all possible *Poker Face* listeners give the song rating of 3.86

| Rate, $x$ | Probability, $p(x)$ |
|-----------|---------------------|
| 1         | $f(5) = 0.43$       |
| 2         | $f(4) = 0.21$       |
| 3         | $f(3) = 0.22$       |
| 4         | $f(2) = 0.07$       |
| 5         | $f(1) = 0.07$       |

**End of lecture 8**

## Section 5.3

# The Binomial Distribution

## The Binomial Experiment:

1. Experiment consists of  $n$  identical trials.
2. Each trial results in either “success” or “failure.”
3. Probability of success,  $p$ , is constant from trial to trial
4. Trials are independent.

**Note:** The probability of failure ( $q$ ) is  $1 - p$  and is constant from trial to trial

If  $x$  is the total number of successes in  $n$  trials of a binomial experiment, then  $x$  is a **binomial random variable**

**Binominal:**  $n$  and  $p$

## Question Purchase at a Discount Store

Suppose that historical sales records indicate that 40 percent of all customers who enter a discount department store make a purchase. What is the probability that the next customer will make a purchase?

- Easily, it is  $p = 0.4$

## Example 5.6 Purchase at a Discount Store

Suppose that historical sales records indicate that 40 percent of all customers who enter a discount department store make a purchase. What is the probability that 2 of the next 3 customers will make a purchase?

- $S$ : the customer makes a purchase
- $F$ : the customer does not make a purchase
- $p = 0.4$

The sample space of the experiment (you can, but you do not have to, make a Tree Diagram to show, next page):

*SSS SSF SFS FSS FFS FSF SFF FFF*

What is the probability that 2 of the 3 three customers will make a purchase?

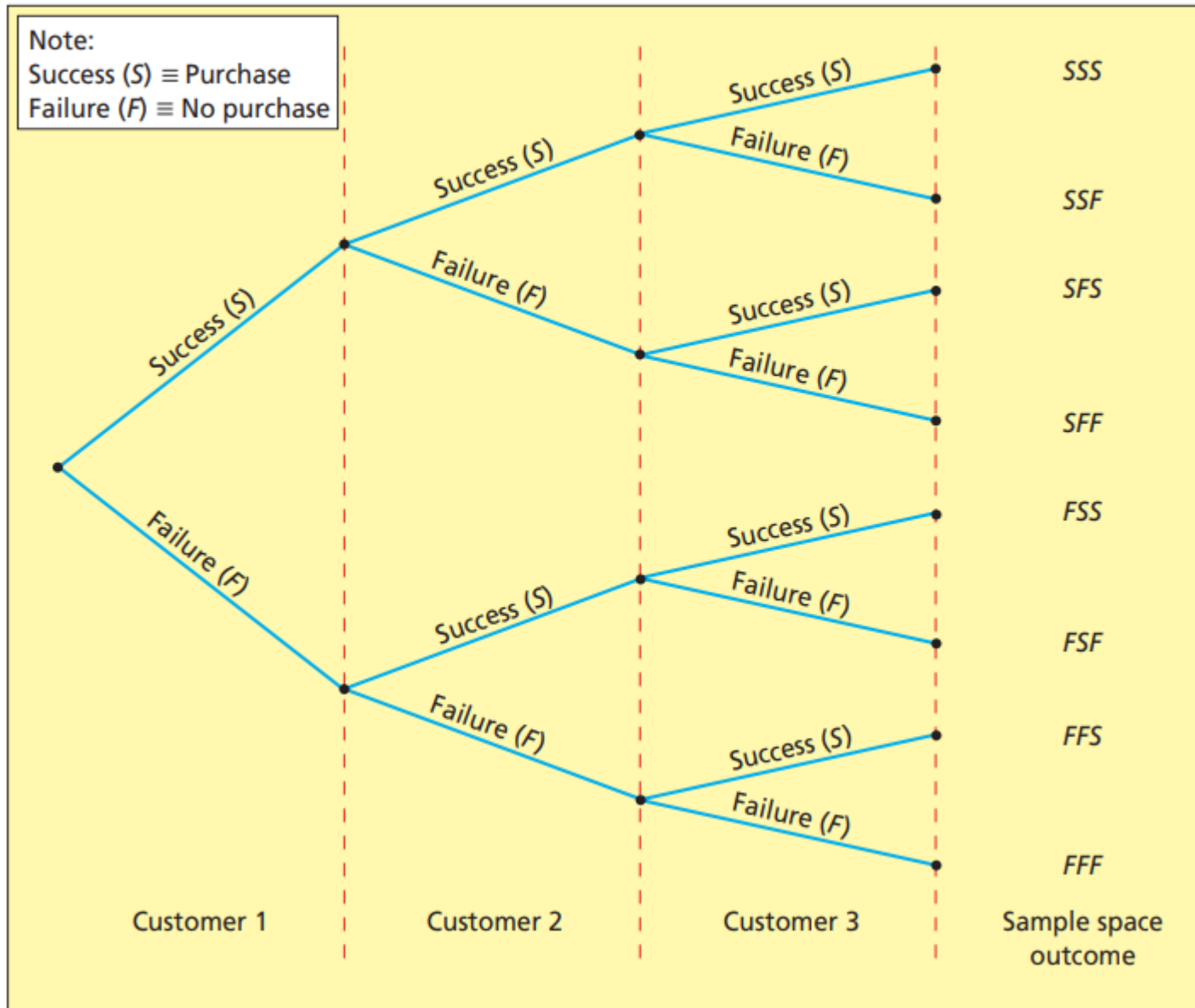
$$P(SSF) = P(S)P(S)P(F) = (0.4)(0.4)(0.6) = (0.4)^2(0.6)$$

$$P(SFS) = P(S)P(F)P(S) = (0.4)(0.6)(0.4) = (0.4)^2(0.6)$$

$$P(FSS) = P(F)P(S)P(S) = (0.6)(0.4)(0.4) = (0.4)^2(0.6)$$

$$P(SSF) + P(SFS) + P(FSS) = 3(0.4)^2(0.6) = 0.288$$

**FIGURE 5.4** A Tree Diagram of Three Customers Making a Purchase Decision



## Question Purchase at a Discount Store

Suppose that historical sales records indicate that 40 percent of all customers who enter a discount department store make a purchase. What is the probability that 3 of the next 5 customers will make a purchase?

- $S$ : the customer makes a purchase
- $F$ : the customer does not make a purchase
- $p = 0.4$

# The Binomial Distribution

For a binomial random variable  $x$ , the probability of  $x$  successes in  $n$  trials is given by the binomial distribution:

$$P(x) = \frac{n!}{x! (n-x)!} p^x q^{n-x}$$

Equation for the binomial distribution

- **Note:**  $n!$  is read as “ $n$  factorial (fak·taw·ree·uhl)” and  $n! = n \times (n-1) \times (n-2) \times \dots \times 1$

➤ For example,  $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

- Also,  $0! = 1$
- Factorials are not defined for negative numbers or fractions.

- What does the equation mean?
  - The equation for the binomial distribution consists of the product of two factors

$$P(x) = \frac{n!}{x! (n-x)!} \times p^x q^{n-x}$$

Number of ways to  
get  $x$  successes and  $(n - x)$  failures in  $n$  trials

The chance of getting  $x$  successes  
and  $(n - x)$  failures in a particular  
arrangement

For previous example (**example 5.6**),  $p = 0.4$ ,  $q = 0.6$ ,  $x = 2$ , number of trials is  $n = 3$ .

$$P(2) = \frac{3!}{(3-2)!} 0.4^2 0.6^{3-2} = 3(0.16)(0.6) = 0.288$$

Number of ways to get 2 purchases and  $(3 - 2) = 1$  non-purchase in 3 trials is 3

## Example 5.7 Purchase at a Discount Store

- $p$  (Purchase) = 0.4
- Want to know the probability that exactly 3 of next 5 customers make purchase

$$P(3) = \frac{5!}{3! (5 - 3)!} 0.4^3 0.6^{5-3} = 10(0.064)(0.36) = 0.2304$$

- A way to avoid hand calculating binomial probabilities is to use **binomial tables**, which have been constructed to give the probability of  $x$  successes in  $n$  trials.
- A table of binomial probabilities is given in Table A.1 (**page 783 of the textbook**).

**TABLE A.1** A Binomial Probability Table:  
Binomial Probabilities ( $n$  between 2 and 6)

| $n = 5$        |       |       |       |       |       |       |       |       |       |       |              |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------|
| $p$            |       |       |       |       |       |       |       |       |       |       |              |
| $x \downarrow$ | .05   | .10   | .15   | .20   | .25   | .30   | .35   | .40   | .45   | .50   |              |
| 0              | .7738 | .5905 | .4437 | .3277 | .2373 | .1681 | .1160 | .0778 | .0503 | .0313 | 5            |
| 1              | .2036 | .3281 | .3915 | .4096 | .3955 | .3602 | .3124 | .2592 | .2059 | .1563 | 4            |
| 2              | .0214 | .0729 | .1382 | .2048 | .2637 | .3087 | .3364 | .3456 | .3369 | .3125 | 3            |
| 3              | .0011 | .0081 | .0244 | .0512 | .0879 | .1323 | .1811 | .2304 | .2757 | .3125 | 2            |
| 4              | .0000 | .0005 | .0022 | .0064 | .0146 | .0284 | .0488 | .0768 | .1128 | .1563 | 1            |
| 5              | .0000 | .0000 | .0001 | .0003 | .0010 | .0024 | .0053 | .0102 | .0185 | .0313 | 0            |
|                | .95   | .90   | .85   | .80   | .75   | .70   | .65   | .60   | .55   | .50   | $x \uparrow$ |

Also for example, the probability that none of four patients will experience nausea (**naw·zee·uh**) is  $p(0) = 0.6561$ , the probability that one of the four patients will experience nausea is  $p(1) = 0.2916$  and so forth. If the value of  $p$  is across the bottom of the table, then we read the values of  $x$  from the right side of the table. As an example, if  $p$  equals 0.70, then the probability of three successes in four trials is  $p(3) = 0.4116$  (we have shaded this probability).

**TABLE 5.4** A Portion of a Binomial Probability Table

**(a) A Table for  $n = 4$  Trials**

|                        |   | ← Values of $p$ (.05 to .50) |       |       |       |       |       |       |       |       |       |   |           |
|------------------------|---|------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|---|-----------|
|                        |   | .05                          | .10   | .15   | .20   | .25   | .30   | .35   | .40   | .45   | .50   |   |           |
| Number of<br>Successes | 0 | .8145                        | .6561 | .5220 | .4096 | .3164 | .2401 | .1785 | .1296 | .0915 | .0625 | 4 |           |
|                        | 1 | .1715                        | .2916 | .3685 | .4096 | .4219 | .4116 | .3845 | .3456 | .2995 | .2500 | 3 |           |
|                        | 2 | .0135                        | .0486 | .0975 | .1536 | .2109 | .2646 | .3105 | .3456 | .3675 | .3750 | 2 | Number of |
|                        | 3 | .0005                        | .0036 | .0115 | .0256 | .0469 | .0756 | .1115 | .1536 | .2005 | .2500 | 1 | Successes |
|                        | 4 | .0000                        | .0001 | .0005 | .0016 | .0039 | .0081 | .0150 | .0256 | .0410 | .0625 | 0 |           |
|                        |   | .95                          | .90   | .85   | .80   | .75   | .70   | .65   | .60   | .55   | .50   | ↑ |           |
|                        |   | Values of $p$ (.50 to .95) → |       |       |       |       |       |       |       |       |       |   |           |

### Example 5.8 The Phe-Mycin Case: Drug Side Effects

- A major drug company has developed a new antibiotic called Phe-Mycin. The company claims that, at most, 10 percent of all patients treated with Phe-Mycin would experience nausea as a side effect of taking the drug.
- Suppose that we randomly select  $n = 4$  patients and treat them with Phe-Mycin. Each patient will either experience nausea (which we arbitrarily call a success) or will not experience nausea (a failure).
- We will assume that  $p$ , the true probability that a patient will experience nausea as a side effect, is 0.10, the maximum value of  $p$  claimed by the drug company.
- Furthermore, it is reasonable to assume that patients' reactions to the drug would be independent of each other.
- Let  $x$  denote the number of patients among the four who will experience nausea as a side effect. It follows that  $x$  is a binomial random variable, which can take on any of the potential values 0, 1, 2, 3, or 4.
- Furthermore, we can calculate the probability associated with each possible value of  $x$  as shown in Table 5.3.
- Because Table 5.3 lists each possible value of  $x$  and also gives the probability of each value, we say that this table gives the **binomial probability distribution of  $x$** .

## Checking the conditions

- A major drug company has developed a new drug.
- 10 percent of patients taking the drug will experience nausea.
- Assume that  $p$  is the probability that a patient will experience nausea. So,  $p = 0.1$ .
- Select  $n = 4$  patients to take the drug.
- Let  $x$  denote the number of patients of four who will experience nausea.
- $x$  is a binomial random variable,  $x = 0$ , or 1, or 2, or 3, or 4.
- We can calculate the probability associated with each possible value of  $x$  as shown in Table 5.3.
- This table gives the binomial probability distribution of  $x$ .

**TABLE 5.3** The Binomial Probability Distribution of  $x$ , the Number of Four Randomly Selected Patients Who Will Experience Nausea as a Side Effect of Being Treated with Phe-Mycin

$x$  (Number Who Experience Nausea)

$$p(x) = \frac{n!}{x! (n - x)!} p^x (1 - p)^{n-x}$$

0

$$p(0) = P(x = 0) = \frac{4!}{0! (4 - 0)!} (.1)^0 (.9)^{4-0} = .6561$$

1

$$p(1) = P(x = 1) = \frac{4!}{1! (4 - 1)!} (.1)^1 (.9)^{4-1} = .2916$$

2

$$p(2) = P(x = 2) = \frac{4!}{2! (4 - 2)!} (.1)^2 (.9)^{4-2} = .0486$$

3

$$p(3) = P(x = 3) = \frac{4!}{3! (4 - 3)!} (.1)^3 (.9)^{4-3} = .0036$$

4

$$p(4) = P(x = 4) = \frac{4!}{4! (4 - 4)!} (.1)^4 (.9)^{4-4} = .0001$$

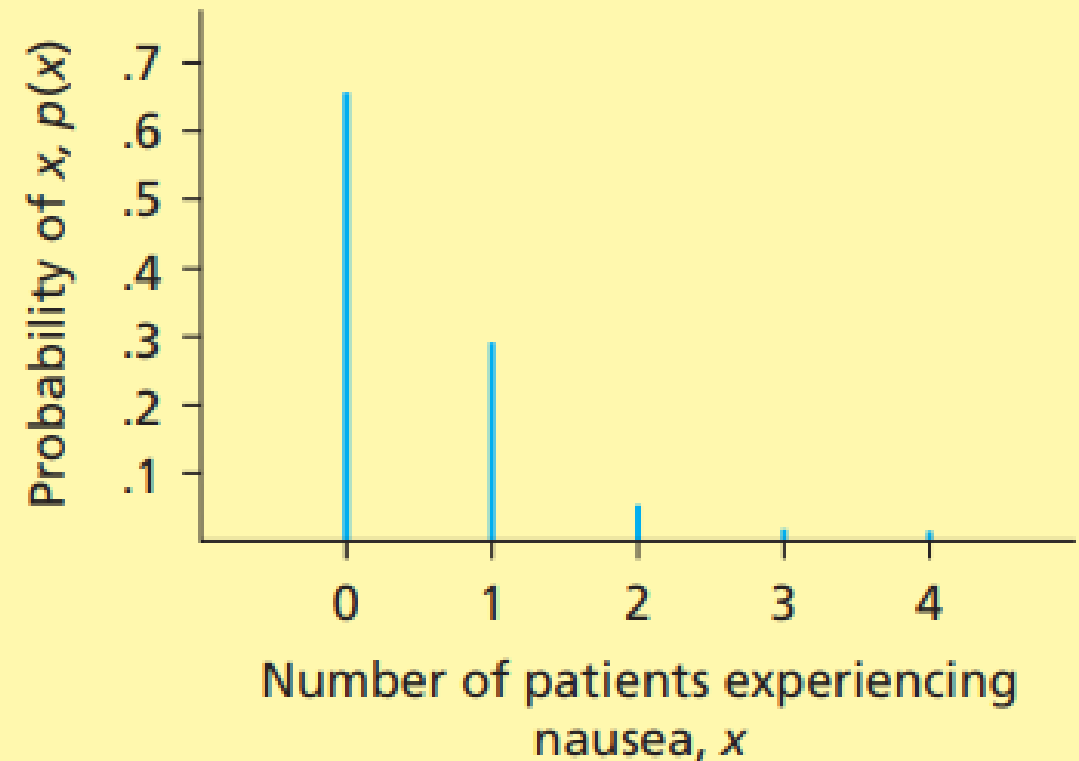
**FIGURE 5.5** The Binomial Probability Distribution with  $p = .10$  and  $n = 4$

(a) Excel output of the binomial distribution

Binomial distribution with  $n = 4$   
and  $p = 0.10$

| $x$ | $P(X = x)$ |
|-----|------------|
| 0   | 0.6561     |
| 1   | 0.2916     |
| 2   | 0.0486     |
| 3   | 0.0036     |
| 4   | 0.0001     |

(b) A graph of the distribution



**x (Number Who  
Experience Nausea)**

$$p(x) = \frac{n!}{x! (n - x)!} p^x (1 - p)^{n-x}$$

0

$$p(0) = P(x = 0) = \frac{4!}{0! (4 - 0)!} (.1)^0 (.9)^{4-0} = .6561$$

1

$$p(1) = P(x = 1) = \frac{4!}{1! (4 - 1)!} (.1)^1 (.9)^{4-1} = .2916$$

2

$$p(2) = P(x = 2) = \frac{4!}{2! (4 - 2)!} (.1)^2 (.9)^{4-2} = .0486$$

3

$$p(3) = P(x = 3) = \frac{4!}{3! (4 - 3)!} (.1)^3 (.9)^{4-3} = .0036$$

4

$$p(4) = P(x = 4) = \frac{4!}{4! (4 - 4)!} (.1)^4 (.9)^{4-4} = .0001$$

In order to interpret these binomial probabilities, consider administering the antibiotic Phe-Mycin to all possible samples of four randomly selected patients. Then, for example,  $P(x = 0) = 0.6561$  says that the probability that none of the four randomly selected patients will experience nausea is 0.6561. In other words, none of the four sampled patients would experience nausea in 65.61 percent of all possible samples.

Furthermore, as another example,  $P(x = 3) = 0.0036$  says that three out of the four sampled patients would experience nausea is only 0.36 percent of all possible samples.

## Example 5.9 The Phe-Mycin Case: Drug side effects

- Suppose that when a sample of randomly selected patients is treated with Phe-Mycin, 3 of the 4 patients experience nausea.
- Because the fraction of patients in the sample that experience nausea is  $3 / 4 = 0.75$ , which is far greater than 0.10, we have some evidence contradicting the assumption that  $p = 0.10$ .
- To evaluate the strength of this evidence, we calculate the probability that at least 3 out of 4 randomly selected patients would experience nausea as a side effect if, in fact,  $p = 0.10$ .

$x$  = number of patients who will experience nausea following treatment with Phe-Mycin out of the 4 patients tested.

Find the probability that at least 3 of the 4 patients treated will experience nausea.

Set  $x = 3$ ,  $n = 4$ ,  $p = 0.1$ , so  $q = 1 - p = 1 - 0.1 = 0.9$ . Then:

$$P(x \geq 3) = P(x = 3 \text{ or } 4) = P(x = 3) + P(x = 4) = 0.0036 + 0.0001 = 0.0037$$

This probability says that, if  $p$  equals 0.10, then in only 0.37 percent of all possible samples of 4 randomly selected patients would at least 3 of the 4 patients experience nausea as a side effect. Because observing such a result is very unlikely, we have very strong evidence that  $p$  does not equal 0.10 and is, in fact, greater than 0.10.

Next, suppose that we consider what our conclusion would have been if only **1 of the 4** randomly selected patients had experienced nausea. Because the sample fraction of patients who experienced nausea is  $1 / 4 = 0.25$ , which is greater than 0.10, we would have some evidence to contradict the assumption that  $p = 0.10$ . To evaluate the strength of this evidence, we calculate the probability that at least 1 out of 4 randomly selected patients would experience nausea as a side effect of being treated with Phe-Mycin if, in fact,  $p = 0.10$ .

$$\begin{aligned} P(x \geq 1) &= P(x = 1 \text{ or } 2 \text{ or } 3 \text{ or } 4) \\ &= P(x = 1) + P(x = 2) + P(x = 3) + P(x = 4) \\ &= 0.2916 + 0.0486 + 0.0036 + 0.0001 = 0.3439 \end{aligned}$$

This means that, if  $p = 0.10$ , the probability of at least 1 of 4 randomly selected patients would experience nausea is 0.3439. Because it is not particularly difficult to believe that a 34.39 percent chance has occurred, we would not have much evidence against the claim that  $p = 0.10$ .

**Rare event** approach to making a statistical inference. The idea of this approach is that if the probability of an observed sample result under a given assumption is small, then we have strong evidence that the assumption is false. Although there are no strict rules, many statisticians judge the probability of an observed sample result to be small if it is less than **0.05**. The logic behind this will be explained more fully in Chapter 9.

In above example, company claims that, at most, 10 percentage of all patients treated with Phe-Mycin would experience nausea. *Is this claim right?*

- Suppose at least 3 of 4 sampled patients *actually* did experience nausea following treatment
  - If  $p = 0.1$  is believed, then there is a chance of only 0.37 percent of observing this result
  - So this is very unlikely!
    - But it actually occurred
  - So, this is very strong evidence that  $p$  does not equal 0.1
    - There is very strong evidence that  $p$  is actually greater than 0.1

### Example 5.10 The ColorSmart-5000 Case: TV Repairs

- The manufacturer of the television set claims that 95 percent of its sets last at least 5 years without requiring a single repair.
- Suppose that we contact  $n = 8$  randomly selected purchasers 5 years after they purchased their sets. Each purchaser's set will have needed no repairs (a success) or will have been repaired at least once (a failure).
- We will assume that  $p$ , the true probability that a purchaser's television set will require no repairs within 5 years, is 0.95, as claimed by the manufacturer. Furthermore, it is reasonable to believe that the repair records of the purchasers' sets are independent of each other.
- Let  $x$  denote the number of the  $n = 8$  randomly selected sets that have lasted at least 5 years without a single repair. Then  $x$  is a binomial random variable that can take on any of the potential values 0, 1, 2, 3, 4, 5, 6, 7, or 8.
- The binomial distribution of  $x$  is listed in Table 5.5.
- Figure 5.6(a) gives the binomial distribution with  $p = 0.95$  and  $n = 8$ . This binomial distribution is graphed in Figure 5.6(b).

**TABLE 5.5** The Binomial Distribution of  $x$ , the Number of Eight ColorSmart-5000 Television Sets That Have Lasted at Least Five Years Without Needing a Single Repair, When  $p = .95$

$x$ , Number of Sets That  
Require No Repairs

$$p(x) = \frac{8!}{x! (8 - x)!} (.95)^x (.05)^{8-x}$$

0

$$p(0) = .0000$$

1

$$p(1) = .0000$$

2

$$p(2) = .0000$$

3

$$p(3) = .0000$$

4

$$p(4) = .0004$$

5

$$p(5) = .0054$$

6

$$p(6) = .0515$$

7

$$p(7) = .2793$$

8

$$p(8) = .6634$$

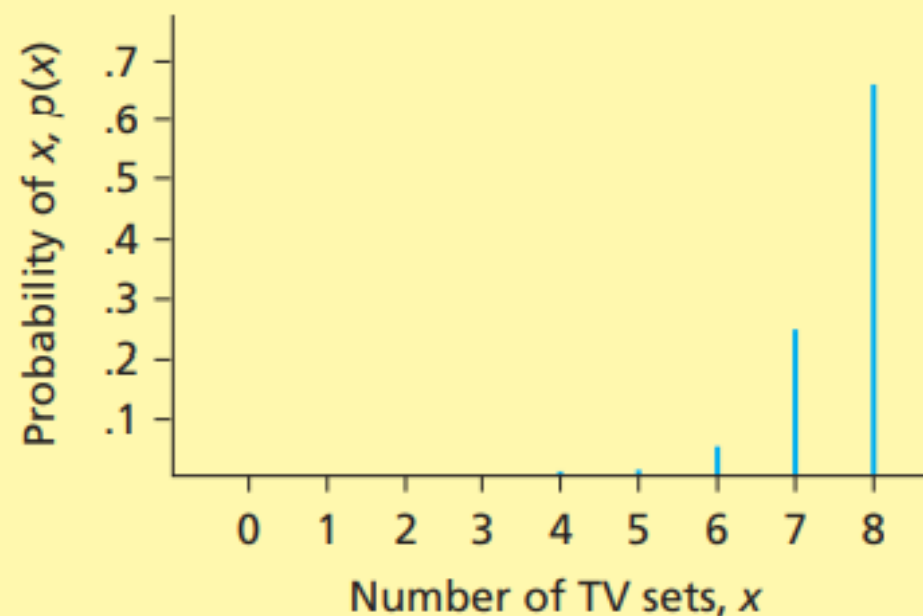
**FIGURE 5.6** The Binomial Probability Distribution with  $p = .95$  and  $n = 8$

(a) MINITAB output of the binomial distribution

Binomial with  $n = 8$  and  $p = 0.95$

| $x$ | $P( X = x )$ |
|-----|--------------|
| 3   | 0.0000       |
| 4   | 0.0004       |
| 5   | 0.0054       |
| 6   | 0.0515       |
| 7   | 0.2793       |
| 8   | 0.6634       |

(b) A graph of the distribution



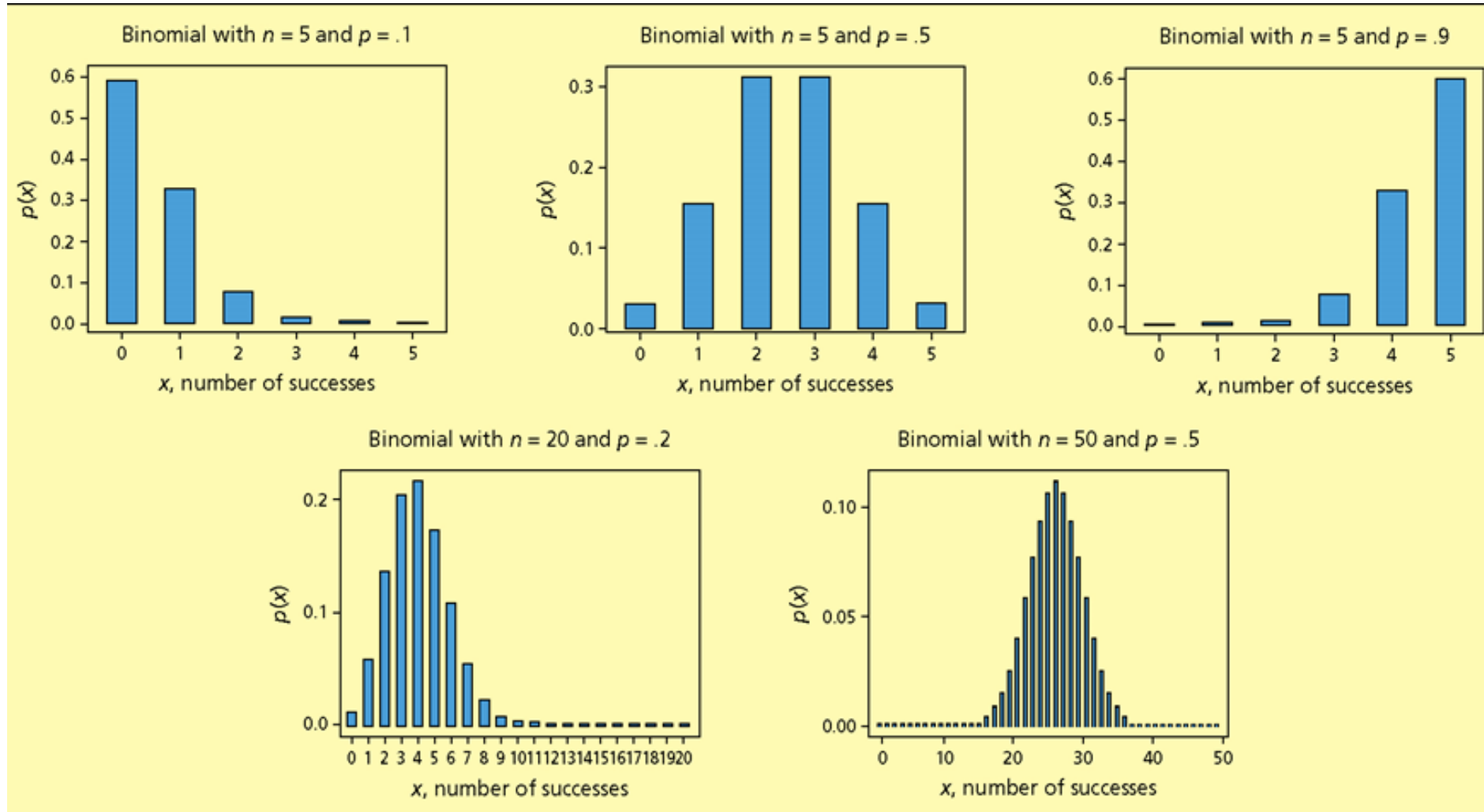
Now, suppose that when we actually contact 8 randomly selected purchasers, we find that 5 out of the 8 television sets owned by these purchasers have lasted at least 5 years without a single repair. Because the sample fraction,  $5 / 8 = 0.625$ , of television sets needing no repairs is less than 0.95, we have some evidence contradicting the manufacturer's claim that  $p = 0.95$ . To evaluate the strength of this evidence, we will calculate the probability that 5 or fewer of the 8 randomly selected televisions would last 5 years without a single repair if, in fact,  $p = 0.95$ . Using the binomial probabilities in Table 5.5, we have

$$\begin{aligned} P(x \leq 5) &= P(x = 5 \text{ or } x = 4 \text{ or } x = 3 \text{ or } x = 2 \text{ or } x = 1 \text{ or } x = 0) \\ &= P(x = 5) + P(x = 4) + P(x = 3) + P(x = 2) + P(x = 1) + P(x = 0) \\ &= .0054 + .0004 + .0000 + .0000 + .0000 + .0000 \\ &= .0058 \end{aligned}$$

This probability says that, if  $p = 0.95$ , then the probability of 5 or fewer of the 8 televisions last 5 years without a single repair is 0.0058. In other words, in only 0.58 percent of all possible samples of 8 randomly selected televisions would 5 or fewer of the 8 televisions last 5 years without a single repair.

Because it is difficult to believe that such a small chance has occurred, we have strong evidence that  $p$  does not equal 0.95, and is, in fact, less than 0.95.

# Several Binomial Distributions



The values of  $n$  and  $p$  are often called the parameters of the binomial distribution. Depending on the parameters, a binomial distribution can be skewed to the right, skewed to the left, or symmetrical.

# Mean and Variance of a Binomial Random Variable

If  $x$  is a binomial random variable with **parameters**  $n$  and  $p$  (so  $q = 1 - p$ ), then

mean  $\mu_x = np$

variance  $\sigma_x^2 = npq$

standard deviation  $\sigma_x = \sqrt{npq}$

where  $n$  is the number of trials,  $p$  is the probability of success on each trial, and  $q = 1 - p$  is the probability of failure on each trial.

## Verification example: Television Manufacturer

**FIGURE 5.6** The Binomial Probability Distribution with  $p = .95$  and  $n = 8$

(a) MINITAB output of the binomial distribution

Binomial with  $n = 8$  and  $p = 0.95$

| x | P( X = x ) |
|---|------------|
| 3 | 0.0000     |
| 4 | 0.0004     |
| 5 | 0.0054     |
| 6 | 0.0515     |
| 7 | 0.2793     |
| 8 | 0.6634     |

From the formulas (Slide #19 and 25):

$$\mu_x = \sum_{all\ x} xP(x)$$

$$\sigma_x^2 = \sum_{Allx} (x - \mu_x)^2 P(x)$$

we get:

$$\mu_x = 4(0.0054) + \dots + 8(0.6634) = 7.6$$

$$\begin{aligned}\sigma_x^2 &= \sum_{Allx} (4 - 7.6)^2(0.004) + \dots + (8 - 7.6)^2 (0.6634) \\ &= 0.38\end{aligned}$$

Given  $p = 0.95$ ,  $q = 0.05$ ,  $n = 8$

$$\mu_x = np = 8(0.95) = 7.6$$

$$\sigma_x^2 = npq = 8(0.95)(0.05) = 0.38$$

Formulas of (Slides #19 and 25) and formulas here (slide 67) yield the same mean  $\mu_x$

In order to interpret  $\mu_x = 7.6$ , suppose that we were to randomly select all possible samples of 8 televisions and record the number of sets in each sample that last 5 years without a repair. If we averaged all of our results, we would find that the average number of sets per sample that last 5 years without a repair is equal to 7.6.

## Other example

- Pat Statsdud is registered in a statistics course and intends to rely on luck to pass the next quiz.
- The quiz consists of 10 multiple choice questions with 5 possible choices for each question, only one of which is the correct answer.
- Pat will guess the answer to each question
- Find the following probabilities
  - Pat gets no answer correct.
  - Pat gets two answer correct?
  - Pat fails the quiz.
  - If all the students in Pat's class intend to guess the answers to the quiz, what is the mean and the standard deviation of the quiz mark?

## Solution

### □ Checking the conditions

- An answer can be either correct or incorrect.
- There is a fixed finite number of trials ( $n = 10$ )
- Each answer is independent of the others. The probability  $p$  of a correct answer (0.20) does not change from question to question.

### □ Determining the binomial probabilities

- Let  $x$  = the number of correct answers

$$P(x = 0) = \frac{10!}{0! (10 - 0)!} (0.20)^0 (0.80)^{10-0} = 0.1074$$

$$P(x = 2) = \frac{10!}{2! (10 - 2)!} (0.20)^2 (0.80)^{10-2} = 0.3020$$

Pat fails the test if the number of correct answers is less than 5, which means less than or equal to 4.

$$\begin{aligned} P(x \leq 4) &= P(0) + P(1) + P(2) + P(3) + P(4) \\ &= 0.1074 + 0.2684 + 0.3020 + 0.2013 + 0.0881 = 0.9672 \end{aligned}$$

The mean and the standard deviation of the quiz mark?

$$\mu = np = 10(0.2) = 2.0$$

$$\sigma = [np(1-p)]^{1/2} = [10(0.2)(0.8)]^{1/2} = 1.26$$

This means that if all the students in Pat's class intend to guess the answers to the quiz, by average, they **only get 2 correct answers**.

## Exercises (if time permits)

**5.19** Suppose that  $x$  is a binomial random variable. Explain what the values of  $x$  represent. That is, how are the values of  $x$  defined?

## Exercises (if time permits)

### 5.21

Suppose that  $x$  is a binomial random variable with  $n = 5$ ,  $p = .3$ , and  $q = .7$ .

- a** Write the binomial formula for this situation and list the possible values of  $x$ .
- b** For each value of  $x$ , calculate  $p(x)$ , and graph the binomial distribution.
- c** Find  $P(x = 3)$ .
- d** Find  $P(x \leq 3)$ .
- e** Find  $P(x < 3)$ .
- f** Find  $P(x \geq 4)$ .
- g** Find  $P(x > 2)$ .
- h** Use the probabilities you computed in part *b* to calculate the mean,  $\mu_x$ , the variance,  $\sigma_x^2$ , and the standard deviation,  $\sigma_x$ , of this binomial distribution. Show that the formulas for  $\mu_x$ ,  $\sigma_x^2$ , and  $\sigma_x$  given in this section give the same results.
- i** Calculate the interval  $[\mu_x \pm 2\sigma_x]$ . Use the probabilities of part *b* to find the probability that  $x$  will be in this interval.

## Answer

$$P(x) = \frac{5!}{x!(5-x)!} 0.3^x 0.7^{5-x}$$

$$x = 0, 1, 2, 3, 4, 5$$

$$P(0, 1, 2, 3, 4, 5) = 0.1681, 0.3602, 0.3087, 0.1323, 0.0284, 0.0024$$

$$P(x = 3) = 0.1323$$

$$P(x \leq 3) = P(x = 0) + P(x = 1) + P(x = 2) + P(x = 3) = 0.9692$$

$$P(x < 3) = P(x = 0) + P(x = 1) + P(x = 2) = 0.837$$

$$P(x \geq 4) = P(x = 4) + P(x = 5) = 0.0308$$

$$P(x > 2) = P(x = 3) + P(x = 4) + P(x = 5) = 0.1631$$

(h)

From the formulas (Slide #19 and 25):

$$\begin{aligned}\mu_x &= \sum_{\text{all } x} xP(x) = 1(0.3602) + 2(0.3087) + 3(0.1323) + 4(0.0284) + 5(0.0024) \\ &= 1.5001\end{aligned}$$

Also the formula given in this section:  $\mu_x = np = 5(0.3) = 1.5$

$$\sigma_x^2 = \sum_{\text{All } x} (x - \mu_x)^2 P(x) = (0 - 1.5)^2 (0.1681) + \cdots (5 - 1.5)^2 (0.0024) = 1.05$$

Also the formula given in this section:  $\sigma_x^2 = npq = 5(0.3)(0.7) = 1.05$

$$\sigma_x = \sqrt{\sigma_x^2} = 1.0247$$

(i)

$$[\mu_x \pm 2(1.0247)] = [1.5 \pm 2.0494] = [-0.5494, 3.5494]$$

$$P(-0.5494 \leq x \leq 3.5494) = P(x = 0) + P(x = 1) + P(x = 2) + P(x = 3) = 0.9692$$

## Exercises (if time permits)

### 5.23

The customer service department for a wholesale electronics outlet claims that 90 percent of all customer complaints are resolved to the satisfaction of the customer. In order to test this claim, a random sample of 15 customers who have filed complaints is selected.

- a** Let  $x$  = the number of sampled customers whose complaints were resolved to the customer's satisfaction. Assuming the claim is true, write the binomial formula for this situation.
- b** Use the binomial tables (see Table A.1, page 783) to find each of the following if we assume that the claim is true:
  - (1)  $P(x \leq 13)$ .
  - (2)  $P(x > 10)$ .
  - (3)  $P(x \geq 14)$ .
  - (4)  $P(9 \leq x \leq 12)$ .
  - (5)  $P(x \leq 9)$ .
- c** Suppose that of the 15 customers selected, 9 have had their complaints resolved satisfactorily. Using part *b*, do you believe the claim of 90 percent satisfaction? Explain.

## Answer

$$P(x) = \frac{15!}{x!(15-x)!} 0.9^x 0.1^{15-x}$$

(b)

$$P(x \leq 13) = 0.2669 + 0.1285 + 0.0428 + 0.0105 + 0.0019 + 0.0003 = 0.4509$$

$$P(x > 10) = 0.9873$$

$$P(x \geq 14) = 0.3432 + 0.2059 = 0.5491$$

$$P(9 \leq x \leq 12) = 0.1837$$

$$P(x \leq 9) = 0.0022$$

(c) Suppose that of the 15 customers selected, 9 have had their complaints resolved satisfactorily. Using part b, do you believe the claim of 90 percent satisfaction? Explain. (No,  $P(x \leq 9)$  is too small)

$n = 15$  $p$  $x \downarrow$ **.05****.10****.15****.20****.25****.30****.35****.40****.45****.50**

0

.4633

.2059

.0874

.0352

.0134

.0047

.0016

.0005

.0001

.0000

15

1

.3658

.3432

.2312

.1319

.0668

.0305

.0126

.0047

.0016

.0005

14

2

.1348

.2669

.2856

.2309

.1559

.0916

.0476

.0219

.0090

.0032

13

3

.0307

.1285

.2184

.2501

.2252

.1700

.1110

.0634

.0318

.0139

12

4

.0049

.0428

.1156

.1876

.2252

.2186

.1792

.1268

.0780

.0417

11

5

.0006

.0105

.0449

.1032

.1651

.2061

.2123

.1859

.1404

.0916

10

6

.0000

.0019

.0132

.0430

.0917

.1472

.1906

.2066

.1914

.1527

9

7

.0000

.0003

.0030

.0138

.0393

.0811

.1319

.1771

.2013

.1964

8

8

.0000

.0000

.0005

.0035

.0131

.0348

.0710

.1181

.1647

.1964

7

9

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## Exercises (if time permits)

### 5.27

The January 1986 mission of the Space Shuttle *Challenger* was the 25th such shuttle mission. It was unsuccessful due to an explosion caused by an O-ring seal failure.

- a** According to NASA, the probability of such a failure in a single mission was  $1/60,000$ . Using this value of  $p$  and assuming all missions are independent, calculate the probability of no mission failures in 25 attempts. Then calculate the probability of at least one mission failure in 25 attempts.
- b** According to a study conducted for the Air Force, the probability of such a failure in a single mission was  $1/35$ . Recalculate the probability of no mission failures in 25 attempts and the probability of at least one mission failure in 25 attempts.
- c** Based on your answers to parts *a* and *b*, which value of  $p$  seems more likely to be true? Explain.
- d** How small must  $p$  be made in order to ensure that the probability of no mission failures in 25 attempts is .999?

## Answer

**(a)**

$$P(x) = \frac{25!}{x! (25 - x)!} \left( \frac{1}{60,000} \right)^x \left( 1 - \frac{1}{60,000} \right)^{25-x}$$

$$P(x = 0) = 0.99958$$

$$P(x \geq 1) = 1 - P(x = 0) = 0.0004166$$

**(b)**

$$P(x) = \frac{25!}{x! (25 - x)!} \left( \frac{1}{35} \right)^x \left( 1 - \frac{1}{35} \right)^{25-x}$$

$$P(x = 0) = 0.4845$$

$$P(x \geq 1) = 1 - P(x = 0) = 0.5155$$

**(c)** The probability (b) is more likely to be true

**(d)**

$$P(x) = \frac{25!}{x! (25 - x)!} (p)^x (1 - p)^{25-x}$$

$$(1 - p)^{25} = 0.999$$

$$(1 - p)^{25} = 0.999$$

$$p = 0.000040019 = \frac{1}{24988}$$

Thank you!